

D467

Exploring Data

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A city transit agency with 1,100 buses wants to know rider satisfaction across all routes. An intern who is a statistics major surveys 400 riders at the downtown hub between 7 and 9 a.m. on weekdays. What is the main problem with the data?

- A. The intern is not qualified to run a survey.
- B. 400 responses is too small to ever be useful.
- C. The agency should survey buses rather than riders.
- D. Sampling bias: weekday rush-hour riders at one hub do not represent all riders.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A clinic database built in-house in 2016 has patients(patient_id, name, dob) and visits(visit_id, patient_id, visit_date). The database runs on a server in the basement. In the visits table, what is patient_id?

- A. A composite key made with visit_date.
- B. A duplicate column that should be deleted.
- C. A foreign key referencing patients.
- D. The primary key of visits.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

In visits(visit_id, patient_id, visit_date), which column must be unique and never null in every row?

- A. visit_date
- B. patient_id
- C. Every column
- D. visit_id

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An analyst at a food distributor with 1,800 SKUs finds customer names like “ Acme Foods” and “Acme Foods ” counted as different customers. The data came from a vendor portal launched in 2022. Which spreadsheet function fixes this?

- A. TRIM
- B. LEN
- C. PROPER
- D. CONCATENATE

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A sales table at a sporting goods chain stores price as text, so an ascending sort returns 100, 25, 9. The table was migrated from an old ERP system in March. Which fixes the sort?

- A. ORDER BY CAST(price AS STRING)
- B. ORDER BY CAST(price AS NUMERIC)
- C. ORDER BY price DESC
- D. ORDER BY TRIM(price)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A text column holds the values '40', '5', and '300'. What does an ascending ORDER BY on that column return?

- A. '300', '40', '5'
- B. '40', '5', '300'
- C. An error, because text cannot be sorted.
- D. '5', '40', '300'

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst at a telecom with a new CEO cleans a 52,000-row customer file, removing 1,200 duplicates and 300 test accounts. Her team uses Google Sheets. The cleaned file has 49,800 rows. What should she conclude?

- A. 2,200 rows are missing.
- B. The file has 700 rows too many.
- C. 700 rows are unaccounted for; find and document them before analysis.
- D. The file is clean; fewer rows means more problems were caught.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A purchasing analyst at a 450-bed hospital uses VLOOKUP to pull unit prices from a price list sorted by vendor name. Items that are not on the list are returning prices of other items. The CFO recently asked for weekly reports. What is the likely cause?

- A. VLOOKUP cannot handle more than 1,000 rows.
- B. The last argument is TRUE or missing; set it to FALSE for an exact match.
- C. The list must be sorted in descending order.
- D. The lookup value must be in the rightmost column.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which formula counts how many orders in C2:C500 are from the West region?

- A. =COUNTA(C2:C500)
- B. =COUNT(C2:C500)
- C. =SUMIF(C2:C500,"West")
- D. =COUNTIF(C2:C500,"West")

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A retailer randomly surveyed 400 of its 90,000 loyalty members and got a margin of error of ± 5 points at 95% confidence. The survey ran during the holiday rush. Leaders want ± 2.5 points. About how many members must they survey?

- A. 200
- B. 1,600
- C. 800
- D. All 90,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A random sample grows from 500 to 2,000. What happens to the margin of error?

- A. It is cut in half.
- B. It falls to one-quarter.
- C. It stays the same because the population has not changed.
- D. It doubles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A fitness app with 2 million users, rated among the top five in its category, wants to share workout data with an insurance partner. Users agreed to data use for “improving the app.” What is the most ethical step?

- A. Share it, since users already accepted the terms.
- B. Remove names and share the rest.
- C. Get explicit consent for the new purpose before sharing, and share only what is needed.
- D. Share it, because top-rated apps commonly do.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A city transit agency with 1,100 buses wants to know rider satisfaction across all routes. An intern who is a statistics major surveys 400 riders at the downtown hub between 7 and 9 a.m. on weekdays. What is the main problem with the data?

Target word: flaw in the sample **Decides:** One hub and rush hour only, for a question about all riders.

A	Camouflage	The intern's background does not create the bias; the design does.
B	Absolute language	400 can be plenty if it is representative; the problem is who was sampled.
C	Adjacent category	The question is about rider satisfaction.
D	Correct	Night, weekend, and outlying-route riders are left out.

2. Answer C Concept 2

A clinic database built in-house in 2016 has patients(patient_id, name, dob) and visits(visit_id, patient_id, visit_date). The database runs on a server in the basement. In the visits table, what is patient_id?

Target word: key type **Decides:** patient_id in visits points to the patients table.

A	Adjacent category	Nothing in the design combines those columns into a key.
B	Inverted operation	Deleting it would break the link between the tables.
C	Correct	It links each visit to a patient and can repeat.
D	Confusion set	visit_id is the primary key; one patient has many visits.

3. Answer D Concept 2 · transfer

In visits(visit_id, patient_id, visit_date), which column must be unique and never null in every row?

Target word: reverse: identify the primary key **Decides:** The primary key uniquely identifies each row.

A	Adjacent category	Many visits can share a date.
B	Confusion set	A foreign key; it repeats for returning patients.
C	Absolute language	Only the primary key must be unique.
D	Correct	Each visit has its own ID.

4. Answer A Concept 3

An analyst at a food distributor with 1,800 SKUs finds customer names like " Acme Foods" and "Acme Foods " counted as different customers. The data came from a vendor portal launched in 2022. Which spreadsheet function fixes this?

Target word: cleaning function **Decides:** Leading and trailing spaces.

A	Correct	Removes the extra spaces so the names match.
B	Confusion set	Shows the length difference but does not fix it.
C	Adjacent category	Fixes capitalization, not spaces.
D	Inverted operation	Joins text together instead of cleaning it.

5. Answer B Concept 4

A sales table at a sporting goods chain stores price as text, so an ascending sort returns 100, 25, 9. The table was migrated from an old ERP system in March. Which fixes the sort?

Target word: fix for sort order **Decides:** Text sorts character by character.

A	Confusion set	It is already text.
B	Correct	Converting to a number sorts 9, 25, 100.
C	Inverted operation	Reverses the text order without fixing the type.
D	Adjacent category	Removes spaces but leaves it as text.

6. Answer A Concept 4 • transfer

A text column holds the values '40', '5', and '300'. What does an ascending ORDER BY on that column return?

Target word: predict text sort **Decides:** Text compares the first character first.

A	Correct	The first characters 3, 4, 5 set the order.
B	Dropped element	The original order, as if no sort ran.
C	Absolute language	Text sorts alphabetically.
D	Confusion set	That is numeric order.

7. Answer C Concept 5

An analyst at a telecom with a new CEO cleans a 52,000-row customer file, removing 1,200 duplicates and 300 test accounts. Her team uses Google Sheets. The cleaned file has 49,800 rows. What should she conclude?

Target word: reconcile counts **Decides:** $52,000 - 1,500 = 50,500$ expected.

A	Wrong stem number	$52,000 - 49,800$ ignores the 1,500 documented removals.
B	Inverted operation	It has 700 fewer rows than expected.
C	Correct	$50,500$ expected – $49,800$ actual = 700 missing.
D	Dropped element	Skips the reconciliation step.

8. Answer B Concept 6

A purchasing analyst at a 450-bed hospital uses VLOOKUP to pull unit prices from a price list sorted by vendor name. Items that are not on the list are returning prices of other items. The CFO recently asked for weekly reports. What is the likely cause?

Target word: VLOOKUP error cause **Decides:** Approximate match returns the nearest lower value.

A	Absolute language	No such limit exists.
B	Correct	Exact match returns #N/A for missing items instead of a neighbor's price.
C	Inverted operation	Sorting does not fix approximate matching.
D	Confusion set	VLOOKUP searches the leftmost column.

9. Answer D Concept 6 · transfer

Which formula counts how many orders in C2:C500 are from the West region?

Target word: conditional count **Decides:** Counting rows that meet one condition.

A	Adjacent category	Counts every non-empty cell, all regions.
B	Dropped element	Drops the condition and counts numbers only.
C	Confusion set	SUMIF adds values; it does not count.
D	Correct	Counts cells equal to West.

10. Answer B Concept 7

A retailer randomly surveyed 400 of its 90,000 loyalty members and got a margin of error of ± 5 points at 95% confidence. The survey ran during the holiday rush. Leaders want ± 2.5 points. About how many members must they survey?

Target word: required sample size **Decides:** Halving the margin takes four times the sample.

A	Inverted operation	A smaller sample widens the margin.
B	Correct	$400 \times 4 = 1,600$.
C	Dropped element	Drops the square root; doubling n cuts the margin only by about 29%.
D	Absolute language	A census is not needed for ± 2.5 points.

11. Answer A Concept 7 · transfer

A random sample grows from 500 to 2,000. What happens to the margin of error?

Target word: effect on margin **Decides:** $4 \times$ the sample $\rightarrow$ margin $\div \sqrt{4}$.

A	Correct	$\sqrt{4} = 2$.
B	Confusion set	Ignores the square root.
C	Adjacent category	The margin depends mainly on sample size.
D	Inverted operation	Bigger samples shrink the margin.

12. Answer C Concept 8

A fitness app with 2 million users, ~~rated among the top five in its category~~, wants to share workout data with an insurance partner. Users agreed to data use for “improving the app.” What is the most ethical step?

Target word: ethical action **Decides:** Sharing with an insurer is a new purpose outside the original consent.

A	Adjacent category	That consent covered a different purpose.
B	Dropped element	Detailed health data can often be re-identified; consent is still missing.
C	Correct	New uses need new, informed consent.
D	Camouflage	Ranking and common practice do not create consent.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Sampling bias	A sample must represent the population the question is about; convenience samples skew results.	Thinking a large sample fixes an unrepresentative one.	Data collected at one place, time, or group to answer a question about everyone.
2	Primary and foreign keys	A primary key uniquely identifies each row; a foreign key points to a primary key in another table and can repeat.	Calling any ID column the primary key.	The same ID column appears in two tables.
3	Text cleaning functions	TRIM removes extra spaces; case functions standardize letters; neither changes meaning.	Reaching for a function that measures or joins text instead of fixing it.	Values that look identical but count separately.
4	Data types and CAST	Numbers stored as text sort and compare as text; convert with CAST before sorting or math.	Changing sort direction to fix a type problem.	An odd sort order like 100, 25, 9.
5	Verifying cleaned data	Reconcile row counts: original – documented removals should equal the cleaned count.	Assuming fewer rows means cleaner data.	Before and after counts plus a list of removals.
6	Lookup and conditional functions	VLOOKUP needs FALSE for an exact match; COUNTIF/SUMIF count or add only rows meeting a condition.	Leaving VLOOKUP on approximate match.	Lookups returning values for items that are not on the list.
7	Sample size and margin of error	Margin of error shrinks with the square root of sample size; halving it takes four times the sample.	Assuming doubling the sample halves the margin.	A target margin of error half the current one.
8	Consent and data ethics	Data may be used only for purposes people agreed to; new uses need new consent.	Treating any accepted terms, or removing names, as blanket permission.	A new partner or purpose for existing personal data.

Course scope (public description): Exploring Data builds proficiency with data, including the organization, preparation, transformation, cleaning, and verification of data. This course examines how to apply critical thinking, spreadsheet, and structured query language techniques to data management and decision-making. Students may simultaneously work toward the Google Data Analytics Professional Certificate and the WGU Business Analysis Certificate. The course D468: Discovering Data, which also includes Google certificate materials, is a prerequisite.